

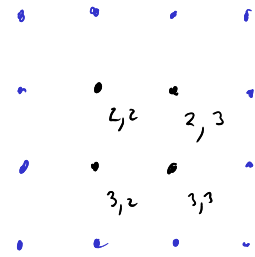
Relaxation Method for Solving Linear Systems of Equations.

Consider the system of equations to solve the Laplace equation,

$$\begin{bmatrix} -4 & 1 & 1 & 0 \\ 1 & -4 & 0 & 1 \\ 1 & 0 & -4 & 1 \\ 0 & 1 & 1 & -4 \end{bmatrix} \begin{bmatrix} u_{22} \\ u_{23} \\ u_{32} \\ u_{33} \end{bmatrix} = \begin{bmatrix} -175 \\ -75 \\ -150 \\ -50 \end{bmatrix}$$

Which correspond to the picture,

We want to solve for the points inside the grid,



These points from Laplace equation are expected to be the average of the 4 nearest neighbors,

$$u_{i,j} = \frac{u_{i+1,j} + u_{i-1,j} + u_{i,j-1} + u_{i,j+1}}{4}$$

$$\bar{u}_{i,j} = \frac{u_{i+1,j} + u_{i-1,j} + u_{i,j-1} + u_{i,j+1}}{4}$$

We want that, $u_{i,j} \approx \bar{u}_{i,j}$

Hence at some iteration t , we move from $u_{i,j}^t$ to $u_{i,j}^{t+1}$

By moving along $(\bar{u}_{i,j}^t - u_{i,j}^t)$ with learning rate ω

Hence
$$u_{i,j}^{t+1} = u_{i,j}^t + \omega (\bar{u}_{i,j}^t - u_{i,j}^t)$$

Hence,
$$u_{i,j}^{t+1} = \omega \bar{u}_{i,j}^t + (1 - \omega) u_{i,j}^t$$

this is the fundamental equation of the relaxation method,

Depending on the value of w we have different convergence,

$0 < w < 1$, Subrelaxation

$w = 1$, Gauss Seidel method.

$1 < w < 2$, Over relaxation

$w > 2$, Divergent.

Hence the method works by evolving, $u_{i,j}^0 \rightarrow u_{i,j}^1 \rightarrow \dots \rightarrow u_{i,j}^t \rightarrow u_{i,j}^{t+1} \dots$

the method has a stopping criteria

$$|u_{i,j}^{t+1} - u_{i,j}^t| < \delta \text{ for all } i, j, \text{ and some } \delta > 0.$$